FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Measles

Tutorial | Week 5 · Measles, vaccines & the newborn | Paediatrics | Dr Nabbale

Measles is a highly contagious airborne viral illness (fever, cough, coryza, conjunctivitis, Koplik spots, then a spreading maculopapular rash) whose deaths come from pneumonia, diarrhoea, malnutrition, eye damage and encephalitis; every case gets age-dosed vitamin A, isolation, nutrition support and active treatment of complications, and two doses of measles-rubella vaccine (9 and 18 months in Uganda) prevent it.

Must know before the ward round

1. WHO case definition: fever + generalised maculopapular rash + cough, coryza (runny nose) or conjunctivitis. Measles is notifiable in Uganda: report suspected cases within 24 hours.
2. Timeline: incubation about 10–12 days to the prodrome and about 14 days (7–21) to the rash; prodrome 2–4 days; Koplik spots 1–2 days before the rash; rash spreads from hairline and behind the ears to the feet over 3 days.
3. Infectious from 4 days before to 4 days after the rash appears (for the whole illness if immunocompromised); airborne spread with secondary attack rates of 90% or more in susceptible contacts.
4. Vitamin A by age: under 6 months 50,000 IU; 6–11 months 100,000 IU; 12 months or older 200,000 IU. UCG 2023: day 1, day 2 and day 14 for every case; WHO: 2 doses 24 h apart, plus a third at 2–4 weeks if there are eye signs.
5. Severe complicated measles (IMCI): any general danger sign, clouding of the cornea, or deep or extensive mouth ulcers → vitamin A, first-dose antibiotics, tetracycline eye ointment and urgent referral.
6. Pneumonia is the commonest cause of measles death; more than half is bacterial (S. pneumoniae, S. aureus, Hib), so every child with measles and pneumonia gets antibiotics.
7. Eyes: measles plus vitamin A deficiency is a leading cause of childhood blindness in Africa. Clean the eyes, apply tetracycline 1% eye ointment, never steroid drops, and pad an ulcerated eye.
8. Fever persisting more than 3–4 days after the rash appears, or a new fever, means a complication (pneumonia, otitis media, encephalitis, TB flare).
9. Encephalitis: acute (about 1 in 1,000 cases, during or soon after the rash); ADEM (post-infectious, weeks later); measles inclusion body encephalitis (immunocompromised, months later); SSPE (fatal, years later).
10. SSPE: mean 7–10 years after measles, highest risk if infected under 2 years; school decline and behaviour change → myoclonic jerks → dementia and death; periodic EEG complexes and high CSF measles antibody.
11. Uganda: MR1 at 9 months and MR2 at 18 months. In 2025, MR1 coverage was 92% but MR2 only 64%; 66 districts had outbreaks (over 10,000 cases, 60 deaths). A national MR campaign for 9–59 month olds ran on 1–5 October 2026.
12. Measles vaccine (live MR) is given even to malnourished and HIV-positive children unless severely immunosuppressed; post-exposure vaccine within 72 hours protects contacts aged 6 months or more.

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In a nutshell

Measles (rubeola) is an acute viral illness caused by the measles virus (a *Morbillivirus*). It is one of the most contagious infections known: one case can infect **12–18 susceptible people**. It begins with fever and the "three Cs" (**c**ough, **c**oryza, **c**onjunctivitis), then **Koplik spots**, then a **maculopapular rash** spreading from the head down.

Measles itself is self-limiting. Children die from what follows, especially **pneumonia, diarrhoea and dehydration, malnutrition, croup, encephalitis and corneal ulceration**. The virus also **suppresses immunity for weeks to months**. The poorest children (malnourished, vitamin A deficient, HIV-positive, unvaccinated) do worst. Uganda is in the middle of repeated outbreaks because too many children miss the **second dose (MR2) at 18 months**.

> **CLINICAL PEARL**
> The one thing never to forget: **every child with measles gets vitamin A, by age, today**, plus a hunt for complications (chest, ears, eyes, mouth, hydration, nutrition) and isolation. And every contact who is not vaccinated gets the vaccine.

Definitions & classification

Term	Definition
Suspected measles (WHO/IDSR case definition)	Any person with **fever** and a **generalised maculopapular (non-vesicular) rash** and **cough, coryza or conjunctivitis**, or any person a clinician suspects has measles
Laboratory-confirmed	Measles **IgM** positive (serum, ideally 3–28 days after rash onset) or measles RNA detected by PCR (throat/nasopharyngeal swab, urine)
Epidemiologically linked	Suspected case linked to a laboratory-confirmed case during an outbreak
Non-severe measles	No danger signs, no complications needing admission
Measles with eye or mouth complications (IMCI)	Pus draining from the eye, or mouth ulcers (not deep or extensive)
Severe complicated measles (IMCI)	**Any general danger sign**, **clouding of the cornea**, or **deep or extensive mouth ulcers** (WHO Pocket Book adds pneumonia, dehydration, stridor, severe malnutrition)
Measles outbreak (Uganda IDSR)	Usually 3 or more confirmed cases in a district within a month (check your current district guidance); a single confirmed case triggers investigation

The stages

Stage	Timing	Features
Incubation	About 10–12 days from exposure to first symptoms (range 7–21 days to rash)	Asymptomatic; virus multiplies in the respiratory tract and lymph nodes

Stage	Timing	Features
Prodrome (catarrhal stage)	2–4 days (up to 7)	High fever (often 39–40.5 °C), **cough** (harsh, "barking"), **coryza**, **conjunctivitis** (red, watery eyes, photophobia), irritability, misery, poor appetite, diarrhoea. **Koplik spots** (white/grey "grains of salt" on a red base on the buccal mucosa opposite the lower molars) appear **1–2 days before the rash** and fade 1–2 days after it
Exanthem (rash stage)	Days 3–7 of illness	**Erythematous maculopapular rash** starting at the **hairline, face and behind the ears**, spreading **down the trunk and limbs over 3 days**, reaching hands and feet. Spots become **confluent** (blotchy). Fever peaks 2–3 days after the rash starts, then falls
Recovery	From about day 4 of the rash	The rash fades in the order it appeared, leaving **brownish staining** and fine **desquamation** (spares palms and soles); in malnourished children, dark peeling "flaky paint" skin. Cough may last 1–2 weeks. Immune suppression persists for weeks to months

EXAM FAVOURITE

"Fever should fall 2–3 days after the rash appears." If the fever **persists beyond day 3–4 of the rash** or returns, look for a **complication**: pneumonia, otitis media, diarrhoea, encephalitis, or TB reactivation.

Epidemiology & Uganda context

- **Global (WHO fact sheet):** measles killed an estimated **95,000 people in 2024**, mostly unvaccinated or under-vaccinated children under 5. Vaccination averted about **59 million deaths between 2000 and 2024**. First-dose coverage was **84% in 2025**, below the 95% needed to stop transmission.
- **Uganda (MoH, 2025–2026):**
  - National measles vaccination coverage rose from 79% (2016) to 90% (2024), still below the **95% target**.
  - In 2025, **MR1 coverage was 92% but MR2 coverage only 64%**. Many caregivers don't know that Uganda gives a second dose at 18 months.
  - In 2025, **66 districts** confirmed outbreaks, with **over 10,163 cases and 60 deaths**; about a third of deaths were in **Karamoja**, where malnutrition worsens outcomes.
  - Among cases, **75% were under 5**, **64% had never received MR**, and 33% had only one dose.
  - Between January and April 2026, 20 more districts confirmed outbreaks. The MoH ran a **nationwide MR campaign on 1–5 October 2026** for **all children aged 9–59 months, regardless of previous vaccination**, integrated with the October Integrated Child Health Days.
- **Who dies?** Children under 5 (especially under 1), the **malnourished** (SAM), **vitamin A deficient**, **HIV-positive** and otherwise immunocompromised children, children in crowded camps (refugee settlements), and those who present late.

UGANDA CONTEXT

Uganda's schedule (UNEPI): MR1 at 9 months and **MR2 at 18 months** (measles-rubella combined vaccine, introduced in the 2019 campaign; MR2 added later). During outbreaks, the MoH vaccinates children from **6 months** (a "zero dose" that does not count towards the routine schedule) and runs campaigns for **9–59 months**. **Measles samples** (serum for IgM) go via the hub system to the **Uganda Virus Research Institute (UVRI), Entebbe**, the national measles laboratory.

Pathophysiology

1. **Entry and spread**: the virus enters through respiratory epithelium and conjunctiva, infecting **alveolar macrophages and dendritic cells** via the **SLAM (CD150) receptor**, which is found on immune cells. It is

carried to **regional lymph nodes** (days 2–4), then a **primary viraemia** spreads it to the reticuloendothelial system.
2. **Secondary viraemia** (around day 7–10) seeds the **skin, respiratory tract, gut, conjunctiva, liver and brain**. The virus then exits through epithelial cells via the **nectin-4** receptor on their basal side, which is why it is shed so efficiently from the airway.
3. **The rash** is largely an **immune reaction** (T cells attacking infected endothelial cells in the skin), so it appears as antibodies and cell-mediated immunity develop. That is why **severely immunocompromised children may have little or no rash** but devastating pneumonia or encephalitis.
4. **Immune suppression**: the virus infects and kills **memory T and B cells** ("**immune amnesia**"), erasing antibody memory against other infections for months to years. This explains the **secondary bacterial pneumonia, diarrhoea, TB reactivation and a negative tuberculin test** after measles.
5. **Vitamin A**: measles **consumes vitamin A** (increased use and urinary loss, reduced intake). Low vitamin A damages epithelial surfaces (cornea, gut, airway) and immunity. Supplementation **restores levels and reduces death and blindness**, which is why it is given even to well-nourished children.
6. **Gut and nutrition**: mucosal damage → diarrhoea, protein-losing enteropathy; with fever, anorexia, mouth ulcers and harmful food taboos (withholding meat, fish or milk) → **acute malnutrition and kwashiorkor**.
7. **Brain**: acute encephalitis (direct infection), **ADEM** (autoimmune demyelination), **MIBE** (direct infection in immunosuppressed) and **SSPE** (a defective virus persisting in neurons for years).

Clinical acumen

Focused history

Question	Why it matters
Fever, cough, runny nose, red eyes: when did each start? When did the rash start and where?	Confirms the classic sequence; the day of rash tells you infectiousness and when fever should settle
Contact with a rash illness, school/church/market, refugee settlement, outbreak district	Epidemiological link; outbreak response
Immunisation card: MR1 (9 months), **MR2 (18 months)**, campaign doses	Unvaccinated or 1 dose = susceptible; vaccine failure is possible but rare after 2 doses
Vitamin A in the last month (routine 6-monthly supplementation, campaign)	Avoid duplicating too-recent doses (UCG: give if not given in last month for measles treatment)
Feeding: breastfeeding, appetite, food taboos, weight loss	Malnutrition risk; counter harmful myths
Diarrhoea, vomiting, ability to drink	Dehydration
Fast or difficult breathing, noisy breathing (stridor)	Pneumonia, croup
Ear pain or discharge	Otitis media
Eye pain, photophobia, discharge, "something white on the eye"	Keratitis, ulceration, xerophthalmia
Mouth pain, refusing feeds	Stomatitis, candidiasis, cancrum oris
Convulsions, drowsiness, behaviour change, weakness	Febrile seizure, encephalitis
HIV status (child and mother), TB contact, chronic cough	Severe/prolonged measles, giant-cell pneumonia; TB flare after measles
Traditional eye or skin remedies	Chemical eye injury, skin infection

Examination

1. **ETAT triage**: airway (stridor), breathing (rate, indrawing, SpO_2), circulation, consciousness, convulsions, dehydration.
2. **Weight, length/height, MUAC, oedema of both feet** → weight-for-height Z-score; **SAM** changes management.
3. **Rash**: distribution, confluence, desquamation, any vesicles (think chickenpox), purpura (think meningococcus), skin infection.
4. **Mouth**: Koplik spots (early), ulcers (how deep and extensive?), thrush, **cancrum oris** (gangrene of the cheek).
5. **Eyes**: conjunctivitis, discharge, **corneal clouding**, ulceration, **Bitot's spots** and dryness (xerophthalmia) → vitamin A deficiency.
6. **Ears**: discharge, tympanic membrane.
7. **Chest**: RR for age, indrawing, crackles, wheeze, **stridor at rest** (severe croup).
8. **Hydration**: sunken eyes, skin pinch, ability to drink.
9. **Neurology**: consciousness, neck stiffness, focal signs.
10. **Lymph nodes, liver, spleen, oral thrush** (HIV clues).

Red flags

RED FLAG

- **General danger signs**: unable to drink/breastfeed, vomiting everything, convulsions, lethargy/unconsciousness
- **Clouding of the cornea** or corneal ulcer (risk of blindness within days)
- **Deep or extensive mouth ulcers**; gangrene of the cheek (**cancrum oris**)
- Chest indrawing, SpO_2 <90%, **stridor at rest**
- Severe dehydration; bloody diarrhoea
- **Severe acute malnutrition** (MUAC <11.5 cm, WHZ <–3, oedema of both feet)
- Infant under 6 months, HIV-positive or immunocompromised child
- Altered consciousness, seizures, new focal signs (encephalitis)

Thinking like a clinician

- **"Rash in a febrile child: is it measles?"** Check the **sequence** (prodrome of 3 Cs for 2–4 days → rash from the head downward) and the **Koplik spots**. Measles children look **miserable**; rubella and roseola children usually look well.
- **Fever falls as the rash appears** → think **roseola** (HHV-6, under 2 years). **Fever persists after the rash** → measles complication.
- **Malnourished child with measles** = high risk: give vitamin A, admit, start nutritional rehabilitation, look hard for pneumonia, diarrhoea and TB.
- **Child in a measles outbreak with cough and fast breathing** = pneumonia (often bacterial): give antibiotics, oxygen if hypoxic.
- **Convulsion during measles**: febrile seizure is possible, but drowsiness or focal signs = **encephalitis**; check glucose and do an LP if safe.
- **Months after measles**: weight loss, cough, fever → **TB** (measles activates TB and makes the tuberculin test falsely negative for weeks).
- **Teen with school decline and myoclonic jerks** → **SSPE**: ask about measles in infancy.
- **Common student mistakes**: forgetting vitamin A or giving the wrong dose for age; prescribing steroid eye drops; admitting a measles case to the general ward without isolation; not checking the vaccination status of

the siblings, ward contacts and staff; not notifying; underestimating mouth ulcers that stop a child eating.

EXAM FAVOURITE

Viva favourite: **"Why do we give vitamin A to a well-nourished child with measles?"** Answer: measles itself causes an acute fall in serum vitamin A (increased use and loss, reduced intake). Two doses restore levels, **reduce mortality (especially in children under 2) and pneumonia**, and **prevent corneal damage and blindness**. WHO and UCG recommend it for **every** case, regardless of nutritional status.

Clinical workup

Step 0: First 5 minutes

1. **Mask and separate**: move the child and caregiver to an **isolation area/room** (airborne precautions; only staff with documented immunity attend; N95 if available).
2. **ETAT**: airway (stridor), breathing (SpO_2, indrawing), circulation, coma, convulsions, dehydration.
3. **Oxygen** if SpO_2 <90%, central cyanosis or severe respiratory distress.
4. **Check glucose** if lethargic, convulsing or not feeding.
5. **Weigh, MUAC, oedema**: identify SAM immediately.
6. **Give vitamin A (first dose) now**, and the first dose of antibiotic if there are danger signs, pneumonia, corneal clouding or deep mouth ulcers.

Step 1: Focused history

- [ ] **Sequence and dates** (fever, 3 Cs, rash onset) → confirms measles; sets isolation end (4 days after rash onset).
- [ ] **Contacts/outbreak** → notification and outbreak response.
- [ ] **MR1 and MR2 status** (card), recent campaigns → susceptibility; catch-up for siblings.
- [ ] **Vitamin A received recently** → avoid exceeding doses.
- [ ] **Feeding, diarrhoea, vomiting** → dehydration, malnutrition.
- [ ] **Breathing, stridor, ear, eye and mouth symptoms** → complications.
- [ ] **Convulsions, consciousness** → encephalitis.
- [ ] **HIV (child and mother), TB contact** → severity, TB flare.
- [ ] **Traditional remedies** (eye drops, herbs) → chemical eye injury.

Step 2: Focused examination

- **General**: temperature, weight, MUAC, WHZ, oedema, hydration, pallor.
- **Skin**: rash stage and distribution, desquamation, secondary infection.
- **Eyes**: conjunctiva, cornea (clouding, ulcer), Bitot's spots, photophobia.
- **Mouth**: Koplik spots, ulcers (depth/extent), thrush, cheek gangrene.
- **ENT**: ear discharge, tonsils.
- **Chest**: RR, indrawing, crackles, wheeze, stridor; SpO_2.
- **Neurology**: AVPU, neck, fontanelle, focal signs.
- **Document negatives**: "cornea clear, mouth ulcers superficial, no stridor, no indrawing, MUAC 13.2 cm, no oedema".

Step 3: Bedside tests

Test	How	What it tells you
SpO_2	Pulse oximeter	<90% → oxygen; pneumonia severity
Blood glucose	Glucometer	Hypoglycaemia in a child not feeding or convulsing
MUAC and weight-for-height	Tape, scale, WHO charts	SAM (MUAC <11.5 cm, WHZ <−3, oedema)
Malaria RDT	Finger prick	Co-infection (fever in Uganda)
Fluorescein staining (if available)	Drop, blue light	Corneal ulcer
HemoCue Hb	Finger prick	Anaemia

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Measles IgM (serum)	First suspected cases in a new area; per surveillance protocol (in a confirmed outbreak, sample only a few cases)	IgM positive (best 3–28 days after rash; may be negative in the first 3 days)	Confirms outbreak, triggers response. Sent through the district surveillance officer to **UVRI**
Throat/nasopharyngeal swab or urine for PCR	Early (within 5 days of rash) with serum	Measles RNA; genotyping	Outbreak tracking. UVRI
Full blood count	Admitted children	Lymphopenia, low platelets (rare), high WBC with bacterial infection	Antibiotics; transfusion
Blood culture	Severe pneumonia, sepsis	*S. pneumoniae*, *S. aureus*, Gram-negatives	Antibiotic choice. RRH/NRH
Chest X-ray	Severe or non-resolving pneumonia, suspected TB	Bronchopneumonia, lobar consolidation, interstitial (giant cell) pneumonia, hilar nodes (TB)	Antibiotics, TB workup. Hospitals
Electrolytes, urea	Severe dehydration, SAM	Hypokalaemia, hyponatraemia	Fluids (ReSoMal in SAM)
HIV test	All admitted with unknown status	Positive	ART, cotrimoxazole
LP	Encephalitis signs	Lymphocytes, raised protein, normal glucose	Excludes bacterial meningitis
TB workup (Xpert on sputum/gastric aspirate/stool)	Persistent cough, weight loss after measles	MTB detected	TB treatment
EEG, CSF and serum measles antibody	Suspected SSPE	Periodic high-voltage complexes; very high CSF measles IgG	Diagnosis of SSPE. Mulago/specialist

Step 5: Putting it together

Model one-line summary

> "A 2-year-old girl, 9.5 kg (weight-for-height Z −2.4, MUAC 12.1 cm, no oedema), who received MR1 but not MR2, presenting on day 3 of a generalised maculopapular rash after 4 days of fever, cough, coryza and conjunctivitis, with fast breathing (RR 52), lower chest indrawing and SpO_2 89%: measles complicated by

> severe pneumonia and moderate acute malnutrition."

Problem list: (1) measles, day 3 of rash (infectious until day 4 after rash onset); (2) severe pneumonia with hypoxaemia; (3) moderate acute malnutrition; (4) incomplete immunisation (MR2); (5) household contacts unvaccinated.

EXAM FAVOURITE

Worked example 1: vitamin A by age.

- 5-month-old: **50,000 IU** day 1 and day 2 (UCG: and day 14). From a 200,000 IU capsule: **¼ capsule** (about 2 drops of a capsule with 8 drops; check the capsule labelling).
- 10-month-old: **100,000 IU** (½ capsule of 200,000, or one 100,000 IU blue capsule).
- 3-year-old: **200,000 IU** (one red capsule). The UCG 2023 schedule is **day 1, day 2 and day 14**; WHO gives the third dose (2–4 weeks later) only if there are **eye signs** of vitamin A deficiency.

EXAM FAVOURITE

Worked example 2: is this rash measles? A 14-month-old has 3 days of high fever and is still feeding; on day 4 the fever disappears and a pink rash appears on the trunk; the child is happy. **Roseola** (HHV-6): the rash follows defervescence, starts on the trunk, and the child looks well. A 4-year-old has 4 days of fever, cough, red eyes and white spots on the buccal mucosa, then a rash from behind the ears spreading to the trunk, and the fever is higher on the second day of the rash: **measles**.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Rubella	Milder, low-grade fever, **post-auricular and suboccipital lymphadenopathy**, pink rash that clears in 3 days; arthralgia in older girls	Rubella IgM (sent with measles samples)
Roseola infantum (HHV-6)	Under 2 years; 3–5 days of high fever, **rash appears as the fever falls**, child well; febrile seizures	Clinical
Scarlet fever	Sore throat, **sandpaper rash**, strawberry tongue, circumoral pallor, peeling of fingers	Throat swab/ASO
Chickenpox	**Vesicles** in crops at different stages, itchy	Clinical
Drug eruption / SJS	New drug (cotrimoxazole, nevirapine, antiepileptics, sulfa), target lesions, mucosal erosions, skin peeling	History; stop the drug
Kawasaki disease	Fever ≥5 days, non-purulent conjunctivitis, red cracked lips, hand/foot changes, cervical node; no cough or coryza	Clinical, echocardiogram
Meningococcal septicaemia	**Non-blanching purpura**, shock	Glass test, blood culture
Dengue / other arboviruses	Travel/outbreak area, rash, myalgia, low platelets	Dengue NS1/IgM
Infectious mononucleosis + amoxicillin	Older child, tonsillitis, rash after amoxicillin	Clinical
Enterovirus (hand, foot and mouth)	Vesicles on hands, feet, mouth	Clinical
Parvovirus B19	"Slapped cheek", lacy rash on limbs	Clinical

Investigations

Measles is a **clinical diagnosis** in most cases (UCG 2023). Laboratory confirmation is for **surveillance**: the first suspected cases in a district, or a sample of cases during an outbreak, using **serum IgM** (with throat swab or urine PCR) sent to UVRI. Investigations on the ward target **complications** (SpO_2, glucose, Hb, chest X-ray, cultures, HIV test, TB workup). The full table is in **Clinical workup, Step 4**.

Management

There is **no specific antiviral** for measles (ribavirin is experimental and not used in Uganda). Management is: **isolate, vitamin A, nutrition and fluids, eye and mouth care, and active treatment of complications.**

A. Every child with measles

1. **Isolate** at home or in an isolation room (airborne precautions) until **4 days after the rash appears** (longer if immunocompromised).
2. **Vitamin A** (table below).
3. **Paracetamol 10–15 mg/kg** every 6 hours for fever or pain (no aspirin).
4. **Fluids and feeding**: continue breastfeeding; frequent small meals; extra fluids; ORS for diarrhoea (with zinc). **Counter food myths**: children with measles **should** eat meat, fish, eggs and milk (UCG 2023).
5. **Eye care**: clean the eyes with clean water/saline 2–3 times a day; **tetracycline 1% eye ointment** if there is discharge (UCG: every 12 h for 5 days; WHO Pocket Book: 3 times a day for 7 days). **Never use steroid eye preparations.**
6. **Mouth care**: clean with salted water 4 times a day; **gentian violet 0.25%** to ulcers (UCG/WHO); nystatin for thrush.
7. **Check immunisation** of the child (MR2 after recovery if due) and **all contacts**.
8. **Notify** the district surveillance focal person (measles is notifiable).
9. **Follow up in 2–3 days** (outpatients), or sooner if danger signs appear.

Vitamin A dosing

Age	Dose	Schedule (UCG 2023)	Schedule (WHO)
<6 months	**50,000 IU** orally	Day 1 (at diagnosis), day 2, day 14	Day 1 and day 2; third dose 2–4 weeks later if eye signs
6–11 months	**100,000 IU** orally	Day 1, day 2, day 14	As above
≥12 months	**200,000 IU** orally	Day 1, day 2, day 14	As above

UGANDA CONTEXT

Three doses or two? UCG 2023 gives **three doses to every measles case (day 1, day 2 and day 14)**. The WHO Pocket Book gives **two doses 24 hours apart** and a **third dose 2–4 weeks later only if there are eye signs** of vitamin A deficiency (and before discharge for admitted children). Follow UCG on Ugandan wards, and say why if asked. Cut the capsule and squeeze the drops into the mouth; don't give the capsule whole to a young child. If the child had a routine vitamin A dose in the past month, discuss with your senior (UCG: give if not given in the last month).

B. Severe complicated measles (IMCI / UCG 2023)

Any **general danger sign**, **clouding of the cornea**, or **deep or extensive mouth ulcers**:

1. Give **vitamin A** (first dose).
2. Give the **first dose of an antibiotic for severe bacterial infection**: UCG: **ampicillin 50 mg/kg IM + gentamicin 7.5 mg/kg IM** (or benzylpenicillin 50,000 IU/kg IM at HC II).
3. **Tetracycline eye ointment** if clouding of the cornea or pus draining from the eye; cover the eye with a pad.
4. **Refer urgently** to hospital; keep warm, breastfeed or give fluids on the way; check glucose.

C. Managing complications in hospital

Complication	Management (doses per kg)
Pneumonia (over half bacterial)	**Fast breathing or chest indrawing, no danger sign** (WHO 2024 pneumonia and diarrhoea guideline): **oral amoxicillin at least 40 mg/kg per dose 12-hourly** for **5 days** (3 days for fast breathing only in low-HIV areas); in measles, have a low threshold to admit. **Severe pneumonia** (any danger sign, stridor or **SpO_2 <90%**): **ampicillin 50 mg/kg IV/IM 6-hourly + gentamicin 7.5 mg/kg IV/IM once daily**, or **ceftriaxone 80–100 mg/kg/day**; add **cloxacillin 50 mg/kg 6-hourly** if staphylococcal (pneumatoceles, empyema, skin sepsis). **Oxygen** to keep SpO_2 ≥90%
Croup (laryngotracheobronchitis)	Supportive care, keep the child calm with the mother, oxygen if hypoxic, avoid unnecessary throat examination. **WHO Pocket Book: do not give steroids for measles croup** (unlike ordinary viral croup, where dexamethasone 0.15–0.6 mg/kg is used). **Nebulised adrenaline** (0.5 mL/kg of 1:1,000, max 5 mL) for stridor at rest as a temporary measure; intubation or tracheostomy for impending obstruction (refer)
Otitis media	**Amoxicillin 40 mg/kg 12-hourly** for 5 days; dry wicking if discharge
Diarrhoea / dehydration	ORS (Plan A/B/C as per dehydration), **zinc** (10 mg/day under 6 months, 20 mg/day from 6 months, for 10–14 days); bloody diarrhoea → ciprofloxacin 15 mg/kg 12-hourly 3 days; persistent diarrhoea → nutrition
Mouth ulcers / stomatitis	Salt-water cleaning, **gentian violet 0.25%**; **severe or smelly ulcers**: **benzylpenicillin 50,000 IU/kg 6-hourly + metronidazole 7.5 mg/kg 8-hourly for 5 days** (WHO PB); NGT feeding if the child cannot eat
Cancrum oris (noma)	Penicillin + metronidazole, NGT feeding, nutritional rehabilitation, wound care; plastic surgery later (high mortality if untreated)
Eye complications	Vitamin A, tetracycline eye ointment, eye pad for ulcers, **ophthalmology referral for corneal ulcer** (atropine 1% drops for pain if ulcerated, per ophthalmologist); never steroids
Severe acute malnutrition	WHO 2023 / UCG SAM protocol: F-75 then F-100 or RUTF, antibiotics, no fluid boluses unless shock (ReSoMal for dehydration), vitamin A, measles vaccine if not already vaccinated
Febrile convulsion	Diazepam 0.5 mg/kg PR (max 10 mg); paracetamol; check glucose
Acute encephalitis	Supportive: airway, seizure control (diazepam then phenobarbital 15–20 mg/kg), fluids, glucose; LP to exclude bacterial meningitis; ceftriaxone until excluded
TB flare	Screen any child with persistent cough or weight loss after measles; treat TB per national guidelines

D. Infection control and outbreak response

Action	Detail
Isolation	Single room or measles ward/bay with doors closed; caregivers stay with the child; isolation for **4 days after rash onset** (whole illness if immunocompromised)
Staff	Only staff with **2 documented doses of MCV or prior measles**; N95 masks if available; hand hygiene
Ward contacts	**Vaccinate all other children aged 6 months or more** in the ward/OPD who are not fully vaccinated (WHO PB); a dose given at 6–8 months does not count, so they still need MR1 at 9 months and MR2 at 18 months
Post-exposure prophylaxis	**MR vaccine within 72 hours** of exposure for susceptible contacts aged 6 months or more; **normal human immunoglobulin within 6 days** for high-risk contacts who cannot be vaccinated (infants under 6 months, pregnant women, immunocompromised), where available (rarely in Uganda)
Notification	Report suspected cases to the district within **24 hours** (IDSR); collect samples for the first cases
Community	Line-list cases, ring or district-wide vaccination (Uganda: 9–59 months, or 6 months to 14 years depending on age pattern), vitamin A, health education

E. Prevention: measles vaccination

Item	Detail
Vaccine	Live attenuated **measles-rubella (MR)** vaccine (MMR in some private clinics)
Uganda schedule	**MR1 at 9 months, MR2 at 18 months**; catch-up for any child missing doses; campaign doses (e.g. October 2026 for 9–59 months) are given regardless of previous status
Dose and route	**0.5 mL subcutaneous** (upper arm)
Efficacy	About **85% after one dose at 9 months** (maternal antibody interferes), **about 95% or more after two doses**
Cold chain	Freeze-dried, reconstitute with the diluent supplied; use within **6 hours** of reconstitution; discard at the end of the session; protect from light
Side effects	Fever and mild rash 7–12 days after vaccination; febrile seizures rarely; ITP very rarely
Contraindications	**Severe immunosuppression** (e.g. advanced HIV with very low CD4, chemotherapy, high-dose steroids), **pregnancy**, previous anaphylaxis to a vaccine component
NOT contraindications	**Malnutrition** (vaccinate on admission with SAM if unvaccinated and 6 months or more, again at discharge per protocol), mild illness, HIV infection without severe immunosuppression (WHO: HIV-positive children may get an extra dose at 6 months), recent exposure to measles
Vaccine programme failures	Missed MR2 (64% coverage in 2025), cold-chain breaks, hard-to-reach communities, vaccine hesitancy and myths

UGANDA CONTEXT

Use every contact. UCG 2023 IMCI says: check the immunisation card at **every** sick-child visit and **give all missed doses on that visit** (unless the child is being referred). A child admitted with pneumonia or malaria who missed MR1 or MR2 should be vaccinated before discharge. The October Integrated Child Health Days also give vitamin A and deworming.

F. Setting-specific plan

Setting	Action
VHT / HC II	Recognise, vitamin A, paracetamol, ORS, TEO, isolate at home, refer severe complicated measles with first-dose antibiotic; report
HC III	As above plus admission for moderate cases (e.g. dehydration, non-severe pneumonia) if isolation is possible; MR vaccination of contacts
HC IV / general hospital	Isolation bay, oxygen, IV antibiotics, ORS/IV fluids, SAM management, sample collection
RRH / NRH	Severe pneumonia (CPAP/oxygen), encephalitis, corneal ulcer (ophthalmology), noma surgery, ICU

Complications

System	Complication	Notes
Respiratory	**Pneumonia** (viral giant-cell, secondary bacterial: *S. pneumoniae*, *S. aureus*, Hib, Gram-negatives), **croup/laryngotracheobronchitis**, bronchiolitis obliterans, bronchiectasis later	**Commonest cause of death**
ENT	**Otitis media** (commonest complication), mastoiditis, sinusitis	
GI	**Diarrhoea and dehydration**, stomatitis, mouth ulcers, **cancrum oris (noma)**, appendicitis, hepatitis	
Nutritional	**Acute malnutrition, kwashiorkor**, growth faltering, vitamin A deficiency	Measles precipitates SAM
Eye	Conjunctivitis, keratitis, **corneal ulceration**, **xerophthalmia/keratomalacia**, **blindness**; harmful traditional remedies	Leading cause of childhood blindness in Africa
Neurological	**Febrile seizures**; **acute encephalitis** (about 1 in 1,000; seizures, coma, permanent brain damage); **ADEM** (post-infectious, days–weeks after rash); **measles inclusion body encephalitis** (immunocompromised, 1–6 months later); **SSPE** (years later)	
Immune	**Immune amnesia**, TB reactivation, false-negative tuberculin test	Lasts weeks to months (or longer)
Haematological	Thrombocytopenia, haemorrhagic ("black") measles, DIC	Rare
Cardiac	Myocarditis, pericarditis	Rare
Pregnancy	Maternal pneumonia, miscarriage, preterm birth, congenital measles	
Death	Case fatality 1–3 per 1,000 in high-income settings; up to 3–6% or more in malnourished, crowded settings	

SSPE in more detail

Feature	Detail
What it is	A slowly progressive, **fatal** brain disease caused by a defective (mutant) measles virus persisting in neurons

Feature	Detail
Risk	About **4–11 per 100,000 measles cases** overall, but much higher (estimates of 1 in 1,000–1,700) when measles occurs **under 2 years** (before the vaccine is effective); not caused by the vaccine
Latency	Mean **7–10 years** after measles
Stages	(1) Behaviour change, **declining school performance**, irritability → (2) **myoclonic jerks** (periodic), seizures, ataxia → (3) rigidity, dementia, autonomic instability → (4) vegetative state, death usually within **1–3 years**
Diagnosis	**EEG: periodic high-voltage complexes** (Radermecker); **CSF: markedly raised measles IgG** (raised CSF:serum antibody index, oligoclonal bands); MRI white-matter changes; fundus: chorioretinitis
Treatment	No cure; isoprinosine/interferon and antiepileptics are palliative
Prevention	**Measles vaccination**, and protecting young infants through high community coverage (herd immunity)

Ward-round & exam pearls

Model case presentation

> "Baby A, a 2-year-old girl from Kassanda district, **weight 9.5 kg (weight-for-age Z –2.6), height 82 cm (height-for-age Z –1.8), weight-for-height Z –2.4, MUAC 12.1 cm, no oedema**, admitted yesterday with 4 days of fever, cough, runny nose and red eyes, followed by a rash that began behind the ears 3 days ago and spread to the feet. Her elder brother had a similar illness 2 weeks ago. She received **BCG, OPV, pentavalent, PCV, rotavirus and MR1 at 9 months, but not MR2** at 18 months; vitamin A and albendazole last given 8 months ago. She is still breastfed with family foods twice a day; the family stopped giving her beans and fish 'because of the measles'. Mother is **HIV negative** (ANC test); the child's status is unknown. On admission: temperature 39.2 °C, **RR 54, lower chest indrawing, SpO_2 88% in air**, HR 150, CRT 2 s, confluent maculopapular rash with early brown staining on the face, bilateral conjunctivitis with clear corneas, superficial mouth ulcers, no stridor, alert. RDT negative. **Assessment: measles complicated by severe pneumonia (hypoxaemic) and moderate acute malnutrition, with incomplete immunisation. Management so far:** isolation, oxygen by nasal prongs (now SpO_2 95% on 1 L/min), **vitamin A 200,000 IU** yesterday and today (day 14 dose due), ampicillin 475 mg IV 6-hourly + gentamicin 70 mg IV daily, paracetamol, salt-water mouth care and gentian violet, tetracycline eye ointment, feeding counselling. **Plan:** continue antibiotics (switch to oral amoxicillin when off oxygen), wean oxygen, HIV test, supplementary feeding (RUSF/MAM programme), **MR2 after recovery**, notify the district, vaccinate unvaccinated siblings, isolation until 4 days after rash onset (ends tomorrow)."

Likely viva questions

1. *Describe the rash of measles.* Erythematous maculopapular, starts at hairline/behind ears on day 3–4 of illness, spreads down over 3 days, becomes confluent, fades in the same order with brown staining and fine desquamation.
2. *What are Koplik spots?* Small white/grey spots on a red base on the buccal mucosa opposite the molars, appearing 1–2 days before the rash; pathognomonic.
3. *Vitamin A doses?* 50,000 / 100,000 / 200,000 IU for <6 months / 6–11 months / ≥12 months; UCG day 1, 2 and 14.
4. *Commonest cause of death?* Pneumonia (often secondary bacterial).
5. *When is a measles patient infectious?* 4 days before to 4 days after rash onset.

6. *What is SSPE?* A fatal progressive encephalitis 7–10 years after measles, with myoclonus, dementia, periodic EEG complexes and high CSF measles antibody.
7. *Uganda's measles vaccine schedule?* MR1 at 9 months, MR2 at 18 months.

OSCE tips: in a counselling station, explain isolation, vitamin A (three doses in Uganda), feeding (no taboos), danger signs (fast breathing, not drinking, eye pain or white spot on the eye, convulsions), and **vaccinating siblings**. In a vaccination station, demonstrate checking the card, the vial vaccine monitor (VVM), reconstitution with the correct diluent, **0.5 mL subcutaneous** injection, discarding reconstituted vaccine after 6 hours, and recording.

Mnemonics & memory aids

- **"3 Cs and a K"**: **C**ough, **C**oryza, **C**onjunctivitis + **K**oplik spots (before the rash).
- **"4 before, 4 after"**: infectious from 4 days before to 4 days after the rash appears.
- **Vitamin A "50-100-200"**: 50,000 IU (<6 months), 100,000 IU (6–11 months), 200,000 IU (≥12 months); **UCG "1-2-14"** (days).
- **Complications "MEASLES"**: **M**alnutrition (and mouth ulcers), **E**ncephalitis (and eye damage), **A**cute otitis media, **S**SPE, **L**aryngotracheobronchitis (croup), **E**nteritis (diarrhoea), **S**econdary bacterial pneumonia.
- **Severe complicated measles "DEM"**: **D**anger sign, **E**ye (corneal clouding), **M**outh (deep or extensive ulcers).
- **MR schedule "9 and 18"**.

High-yield summary

Numbers & doses to know

Item	Value
Incubation	About **10–12 days** to prodrome; about **14 days** (7–21) to rash
Prodrome	**2–4 days** (3 Cs, Koplik spots 1–2 days before rash)
Infectious period	**4 days before to 4 days after** rash onset
Isolation	Until **4 days after rash onset** (whole illness if immunocompromised)
Secondary attack rate	**≥90%** in susceptible household contacts; R0 12–18
Vitamin A	**50,000 IU** (<6 m), **100,000 IU** (6–11 m), **200,000 IU** (≥12 m)
Vitamin A schedule	**UCG: day 1, 2, 14**; WHO: day 1 and 2 (+ 3rd at 2–4 weeks if eye signs)
Tetracycline eye ointment	1%, UCG every 12 h × 5 days (WHO 3 times a day × 7 days)
Gentian violet	**0.25%** for mouth ulcers
Severe mouth ulcers	Benzylpenicillin 50,000 IU/kg 6-hourly + metronidazole 7.5 mg/kg 8-hourly × 5 days
Pre-referral antibiotics (UCG)	Ampicillin 50 mg/kg IM + gentamicin 7.5 mg/kg IM
Post-exposure vaccine	Within **72 hours**, from **6 months**
Immunoglobulin	Within **6 days** for high-risk contacts
MR schedule (Uganda)	**9 and 18 months**; outbreak dose from 6 months
MR dose	**0.5 mL subcutaneous**; discard reconstituted vaccine after **6 hours**
Efficacy	~85% (1 dose at 9 m), ~95%+ (2 doses)
Herd immunity threshold	**95%** coverage

Item	Value
Acute encephalitis	About **1 per 1,000** cases
SSPE	**4–11 per 100,000** (higher if infected <2 years); onset 7–10 years later
Uganda 2025	MR1 **92%**, MR2 **64%**; **66 districts** with outbreaks; >10,000 cases, 60 deaths

Classic exam traps

- **"Vitamin A only for malnourished children."** Wrong: **every** child with measles.
- **"Steroid eye drops for measles conjunctivitis."** Wrong: steroids worsen corneal infection; use tetracycline ointment.
- **"Measles is over once the rash appears."** Wrong: the dangerous complications come with or after the rash; persistent fever means a complication.
- **"Malnutrition is a contraindication to measles vaccine."** Wrong: malnourished children need it most.
- **"A dose at 6 months in an outbreak counts as MR1."** Wrong: still give MR1 at 9 months and MR2 at 18 months.
- **"Admit to the general ward."** Wrong: isolate; vaccinate other children on the ward.
- **"Stop meat and fish during measles."** Wrong (and a common local myth): feed more, not less.
- **"Antibiotics treat measles."** Wrong: antibiotics treat complications (pneumonia, otitis, severe mouth ulcers), not the virus.

Questions seniors ask

What makes this measles "severe complicated"? Any general danger sign, clouding of the cornea, or deep or extensive mouth ulcers (IMCI); WHO PB adds pneumonia, dehydration, stridor and severe malnutrition.

Why is pneumonia so common and dangerous in measles? The virus damages the respiratory epithelium and causes immune suppression (immune amnesia), so secondary bacterial pneumonia (pneumococcus, *S. aureus*) follows; over half of measles pneumonia is bacterial.

How do you protect the other children on this ward? Isolate the case, check the vaccination status of every child aged 6 months or more and vaccinate within 72 hours of exposure, check staff immunity, and notify the district.

Her fever has continued for 5 days after the rash. What will you look for? Pneumonia, otitis media, mouth infection, diarrhoea with dehydration, encephalitis, bacteraemia, and TB reactivation; examine the chest, ears, mouth and eyes and check SpO_2.

Why does Uganda still have measles outbreaks when MR1 coverage is over 90%? MR1 at 9 months protects only about 85% of children, and MR2 coverage was only 64% in 2025; zero-dose children cluster in hard-to-reach areas, and measles needs about 95% immunity to stop spreading.

A teenager has declining school performance and jerks. Why ask about measles? SSPE: measles in infancy can lead to fatal encephalitis 7–10 years later; check EEG and CSF measles antibody.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	WHO/IDSR case definition of suspected measles?	**Fever + generalised maculopapular (non-vesicular) rash + cough, coryza or conjunctivitis** (or any case a clinician suspects).
2	What are Koplik spots and when do they appear?	Small **white/grey 'grains of salt' on a red base** on the buccal mucosa opposite the lower molars; appear **1–2 days before the rash** and fade 1–2 days after it. Pathognomonic.
3	Stages of measles?	**Incubation** (~10–12 days to symptoms) → **prodrome** (2–4 days: fever, 3 Cs, Koplik spots) → **exanthem** (rash from hairline/behind ears spreading down over 3 days) → **recovery** (brown staining, fine desquamation).
4	Infectious period of measles?	From **4 days before to 4 days after rash onset** (whole illness if immunocompromised). Isolate until 4 days after the rash appears.
5	Vitamin A doses for measles by age?	**<6 months: 50,000 IU; 6–11 months: 100,000 IU; ≥12 months: 200,000 IU**, orally.
6	Vitamin A schedule: UCG 2023 vs WHO?	**UCG 2023: day 1, day 2 and day 14** for every case. **WHO: day 1 and day 2**, plus a third dose 2–4 weeks later if there are eye signs of vitamin A deficiency.
7	Why give vitamin A to every child with measles?	Measles causes an acute **fall in vitamin A** (use, loss, poor intake); supplementation **reduces mortality** (especially under 2s), **pneumonia** and **corneal damage/blindness**, even in well-nourished children.
8	Severe complicated measles (IMCI)?	Measles now or in the last 3 months with **any general danger sign**, **clouding of the cornea**, or **deep or extensive mouth ulcers** → vitamin A, first-dose antibiotic (ampicillin + gentamicin), TEO, urgent referral.
9	Commonest cause of death in measles?	**Pneumonia**, often secondary bacterial (*S. pneumoniae*, *S. aureus*, Hib); others: diarrhoea/dehydration, malnutrition, croup, encephalitis.
10	Commonest complication of measles?	**Otitis media** (then pneumonia, diarrhoea, croup).
11	Eye care in measles?	Clean eyes 2–3 times daily; **tetracycline 1% eye ointment** if discharge (UCG 12-hourly × 5 days; WHO 3 times daily × 7 days); pad an ulcerated eye; refer corneal ulcers. **Never steroid eye drops.**
12	Mouth ulcers in measles: treatment?	Salt-water mouth cleaning, **gentian violet 0.25%**; severe or smelly ulcers: **benzylpenicillin 50,000 IU/kg 6-hourly + metronidazole 7.5 mg/kg 8-hourly for 5 days**; NGT if not eating.
13	What is cancrum oris (noma)?	Rapidly spreading **gangrene of the cheek and mouth** in malnourished children after measles or other infections; treat with penicillin + metronidazole, feeding and nutrition, wound care; reconstructive surgery later.
14	What does fever persisting >3–4 days after the rash suggest?	A **complication**: pneumonia, otitis media, diarrhoea, encephalitis, bacteraemia or TB reactivation.
15	Neurological complications of measles and their timing?	**Acute encephalitis** (during/soon after rash, ~1/1,000); **ADEM** (days–weeks after); **measles inclusion body encephalitis** (immunocompromised, 1–6 months); **SSPE** (7–10 years later).

#	Question	Answer
16	SSPE: key features?	Fatal persistent measles brain infection, mean **7–10 years** after measles (risk highest if infected **under 2 years**); school decline/behaviour change → **myoclonic jerks** → dementia → death in 1–3 years; **periodic EEG complexes**, **high CSF measles IgG**. Prevented by vaccination.
17	What is 'immune amnesia'?	Measles virus kills **memory B and T cells**, erasing immunity to other infections for months to years: more pneumonia, diarrhoea, **TB reactivation**, and a false-negative tuberculin test.
18	Uganda measles vaccine schedule?	**MR1 at 9 months and MR2 at 18 months** (measles-rubella); outbreak doses from 6 months do not count towards the routine schedule.
19	MR vaccine: dose, route and reconstitution rule?	**0.5 mL subcutaneous** (upper arm); reconstitute only with the supplied diluent; **discard 6 hours** after reconstitution or at the end of the session.
20	Post-exposure prophylaxis for measles contacts?	**MR vaccine within 72 hours** for susceptible contacts **aged 6 months or more**; **immunoglobulin within 6 days** for high-risk contacts who cannot be vaccinated (infants <6 months, pregnant, immunocompromised), where available.
21	Is malnutrition a contraindication to measles vaccine?	**No** — malnourished children are at highest risk and should be vaccinated (e.g. on admission with SAM if unvaccinated and aged 6 months or more). Contraindications: severe immunosuppression, pregnancy, anaphylaxis.
22	Uganda measles situation 2025–2026 (MoH)?	2025: **MR1 92%, MR2 64%**; **66 districts** with outbreaks; **>10,000 cases and 60 deaths** (a third in Karamoja). National MR campaign **1–5 October 2026** for **9–59 months**, regardless of previous doses.
23	How is measles confirmed for surveillance in Uganda?	**Serum measles IgM** (best 3–28 days after rash) ± throat swab/urine PCR, sent via the district to **UVRI, Entebbe**. In a confirmed outbreak, further cases are epidemiologically linked.
24	Measles vs roseola?	**Roseola** (HHV-6, under 2): high fever for 3–5 days, **rash appears as the fever falls**, child well. **Measles**: 3 Cs, Koplik spots, child miserable, fever peaks after the rash starts.
25	Measles vs rubella?	**Rubella**: mild, low fever, **post-auricular/suboccipital lymphadenopathy**, rash fades in 3 days, arthralgia; danger is congenital rubella. **Measles**: high fever, 3 Cs, Koplik spots, confluent rash lasting about a week.
26	Measles croup: steroids?	**WHO Pocket Book: do not give steroids for measles croup.** Supportive care, oxygen, nebulised adrenaline for stridor at rest, airway intervention if obstruction.
27	Common feeding myth in measles?	Withholding **meat, fish, eggs or milk**. UCG 2023: counter it — increase fluids and nutrition; continue breastfeeding.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-year-old girl (12 kg, MUAC 13.6 cm) has had fever, cough, runny nose and red eyes for 4 days, and a maculopapular rash that began behind the ears yesterday. She is alert, drinking well, RR 30, SpO2 97%, corneas clear, a few superficial mouth ulcers, no ear discharge. She received MR1 but not MR2. What is the MOST appropriate management?

A Admit for IV ampicillin and gentamicin

B Give vitamin A 100,000 IU once and discharge

C Vitamin A 200,000 IU today, tomorrow and on day 14, paracetamol, mouth care with gentian violet, isolate at home, and review in 2–3 days

D Prescribe oral ribavirin

E Prescribe prednisolone eye drops for the conjunctivitis

2. A 10-month-old boy (7.4 kg) with measles is seen at a HC II. He is lethargic and unable to breastfeed, and the left cornea looks cloudy. What should the health worker do?

A Give vitamin A 100,000 IU, ampicillin 50 mg/kg IM plus gentamicin 7.5 mg/kg IM, tetracycline eye ointment to both eyes, and refer urgently

B Give vitamin A 200,000 IU and send home with eye ointment

C Apply a steroid eye ointment and review in 3 days

D Refer without any treatment to avoid delay

E Give oral amoxicillin and paracetamol, review tomorrow

3. A 2-year-old girl (9.5 kg, weight-for-height Z −2.4) is on day 3 of a measles rash. She has RR 54, lower chest indrawing and SpO2 88% in air; she is alert and drinking. What is the BEST management of her chest?

A Oral amoxicillin as an outpatient for 3 days

B No antibiotics because measles is viral

C Nebulised salbutamol and discharge

D Oral prednisolone for viral pneumonitis

E Admit; oxygen to keep SpO2 at least 90%, IV ampicillin 475 mg 6-hourly plus gentamicin 70 mg once daily

4. A child with measles is admitted to a general paediatric ward at a HC IV. In the same bay are a 7-month-old with pneumonia (no measles vaccine yet) and a 3-year-old with malaria who received MR1 and MR2. What is the MOST appropriate infection-control step for the contacts?

A Give the 7-month-old immunoglobulin only; no action for the 3-year-old

B Move the measles case to isolation and give the 7-month-old MR vaccine within 72 hours (then MR1 at 9 months and MR2 at 18 months as scheduled)

C Give MR vaccine to both children and count the 7-month-old's dose as MR1

D Discharge all three children immediately

E No action: infants under 9 months cannot be vaccinated

5. A 4-year-old boy (14 kg) had measles 6 days ago. The rash is fading with brown staining, but his fever (38.9 °C) has persisted, and today he cries holding his right ear, which is discharging pus. RR 28, chest clear. Which is the likeliest diagnosis?

A Subacute sclerosing panencephalitis
B Normal course of measles
C Measles inclusion body encephalitis
D Acute otitis media complicating measles
E Kawasaki disease

6. A 5-month-old girl (6 kg) is diagnosed with measles during an outbreak in Kikuube district. Which vitamin A regimen follows UCG 2023?

A 200,000 IU on day 1 and day 2
B 100,000 IU once
C 50,000 IU on day 1, day 2 and day 14
D No vitamin A under 6 months
E 25,000 IU daily for 14 days

7. A 2-year-old (10 kg) is on day 5 of measles when she becomes drowsy and has two generalised seizures. Temperature 38.2 °C, glucose 4.9 mmol/L, Blantyre 3, no neck stiffness, RDT negative. Which is the likeliest explanation and next step?

A Acute measles encephalitis: control seizures, supportive care, and do an LP (if safe) with ceftriaxone until bacterial meningitis is excluded
B Simple febrile convulsion: reassure and discharge
C SSPE: start isoprinosine
D Hypoglycaemia: give 10% dextrose 5 mL/kg
E Measles croup: give nebulised adrenaline

8. A 14-year-old boy is brought to Mulago neurology clinic with 6 months of declining school grades, personality change and, recently, repetitive jerks of his arms every few seconds. His mother says he had measles at 10 months, before he was vaccinated. Which investigation will MOST support the likely diagnosis?

A Serum measles IgM
B Malaria blood slide
C CT scan for neurocysticercosis
D Thyroid function tests
E EEG showing periodic complexes and CSF measles IgG antibody titre

9. Which statement about the measles-rubella vaccine in Uganda is CORRECT?

A It is given intramuscularly at 6 and 10 weeks
B It is given as 0.5 mL subcutaneously at 9 and 18 months, and reconstituted vaccine is discarded after 6 hours
C It is contraindicated in children with moderate acute malnutrition
D One dose at 9 months gives 99% protection
E It can cause SSPE

10. A 3-year-old (9 kg, MUAC 11.2 cm, bilateral pitting oedema of the feet) is admitted from Kotido with measles, watery diarrhoea and some dehydration. He is alert, RR 34, SpO2 96%. Which management is MOST appropriate?

A 0.9% saline 20 mL/kg bolus over 15 minutes

B Outpatient care with RUTF only

C Withhold vitamin A because of the oedema

D Vitamin A, ReSoMal for dehydration, F-75 feeds, antibiotics per the SAM protocol, isolation and measles complications care

E Delay feeding until the diarrhoea stops

11. A 14-month-old boy (9 kg) had high fever for 3 days but was playing between fevers. On day 4 the fever suddenly settled and a pink rash appeared on his trunk. He has no cough, coryza or conjunctivitis. Which is the likeliest diagnosis?

A Measles

B Rubella

C Roseola infantum (HHV-6)

D Scarlet fever

E Meningococcal septicaemia

12. A 6-year-old (17 kg) recovered from measles 6 weeks ago. Since then she has had cough, night sweats and weight loss of 1.5 kg. Her grandfather has a chronic cough. A tuberculin skin test is negative. What is the BEST interpretation?

A TB is still likely: measles suppresses cell-mediated immunity, reactivates TB and causes false-negative tuberculin tests; do Xpert on sputum/gastric aspirate and a chest X-ray

B A negative tuberculin test excludes TB

C This is post-measles bronchiolitis obliterans; no further tests needed

D Give another course of vitamin A only

E The cough is a normal post-measles cough; reassure

13. During the October 2026 national MR campaign, a mother brings her 3-year-old who received MR1 and MR2 on schedule. She asks whether he needs the campaign dose. What is the correct advice?

A No: two documented doses mean he must not be vaccinated again

B Only if a measles IgG test is negative

C Only if he has had measles before

D No: campaigns are only for children under 9 months

E Yes: the campaign targets all children aged 9–59 months regardless of previous vaccination status; an extra dose is safe

14. A 2-year-old (10 kg) on day 4 of measles has a loud barking cough and stridor at rest, with suprasternal recession, SpO2 93% in air. He is alert. What is the MOST appropriate immediate treatment?

A IV dexamethasone 0.6 mg/kg and discharge

B Keep him calm with his mother, oxygen, nebulised adrenaline 5 mL of 1:1,000, and close monitoring with a plan for airway support if he worsens

C Examine the throat with a tongue depressor to look for a foreign body

D Salbutamol nebuliser and oral amoxicillin

E Lie him flat and sedate him

15. Which feature BEST distinguishes measles from rubella in a febrile child with a rash?

- A Presence of a maculopapular rash
- B Fever
- C Rash starting on the face
- D Koplik spots with a 3–4-day prodrome of cough, coryza and conjunctivitis
- E Lymphadenopathy behind the ears

Answers & explanations

1. C · Vitamin A 200,000 IU today, tomorrow and on day 14, paracetamol, mouth care with gentian violet, isolate at home, and review in 2–3 days

C is correct: non-severe measles with superficial mouth ulcers is managed at home (UCG 2023): **vitamin A 200,000 IU (≥12 months) on day 1, 2 and 14**, paracetamol, salt-water mouth care and **gentian violet 0.25%**, isolation until 4 days after rash onset, good feeding, and review. A is wrong: she has no danger signs, pneumonia or other indication for IV antibiotics. B is wrong: 100,000 IU is the dose for 6–11 months, and one dose is not enough. D is wrong: ribavirin is experimental and not used. E is wrong: steroid eye drops can worsen corneal infection; use tetracycline ointment if there is discharge.

2. A · Give vitamin A 100,000 IU, ampicillin 50 mg/kg IM plus gentamicin 7.5 mg/kg IM, tetracycline eye ointment to both eyes, and refer urgently

A is correct: lethargy and inability to breastfeed (general danger signs) and corneal clouding = **severe complicated measles** (IMCI/UCG 2023): give **vitamin A (100,000 IU for 6–11 months)**, a **first dose of ampicillin + gentamicin IM**, **tetracycline eye ointment**, and **refer urgently**. B is wrong: 200,000 IU is for ≥12 months, and he needs referral. C is wrong: steroids risk corneal perforation. D is wrong: pre-referral treatment (vitamin A, antibiotics, TEO) is essential and quick. E is wrong: a child with danger signs needs hospital care.

3. E · Admit; oxygen to keep SpO2 at least 90%, IV ampicillin 475 mg 6-hourly plus gentamicin 70 mg once daily

E is correct: hypoxaemia (SpO2 <90%) makes this **severe pneumonia**, and measles pneumonia is often bacterial; admit, give **oxygen** and **ampicillin 50 mg/kg (475 mg) 6-hourly + gentamicin 7.5 mg/kg (~70 mg) daily** (or ceftriaxone), with nutritional support. A is wrong: WHO 2024 outpatient oral amoxicillin applies only to pneumonia **without** danger signs or hypoxaemia. B is wrong: over half of measles pneumonia is bacterial. C is wrong: there is no wheeze; bronchodilators do not treat pneumonia or hypoxaemia. D is wrong: steroids are not indicated and may worsen infection.

4. B · Move the measles case to isolation and give the 7-month-old MR vaccine within 72 hours (then MR1 at 9 months and MR2 at 18 months as scheduled)

B is correct: isolate the case (airborne spread) and give **post-exposure MR vaccine within 72 hours to susceptible contacts aged 6 months or more**; a dose before 9 months does **not** count, so the infant still needs MR1 at 9 months and MR2 at 18 months. The fully vaccinated 3-year-old does not need another dose. A is wrong: vaccine is the preferred PEP from 6 months; immunoglobulin is for those who cannot be vaccinated (e.g. <6 months). C is wrong: the early dose doesn't count as MR1, and the 3-year-old is already protected by two doses. D is wrong: the children are sick and need care. E is wrong: vaccination from 6 months is recommended in outbreaks and after exposure.

5. D · Acute otitis media complicating measles

D is correct: fever persisting beyond day 3–4 of the rash signals a complication, and ear pain with purulent discharge is **acute otitis media**, the **commonest complication** of measles; treat with **amoxicillin 40 mg/kg 12-hourly for 5 days** and dry wicking. A is wrong: SSPE occurs years later. B is wrong: fever should settle 2–3 days after the rash appears. C is wrong: MIBE affects immunocompromised children months later with seizures and coma. E is wrong: Kawasaki disease lacks cough/coryza and has lip, hand and node changes; this child has a clear measles history.

6. C · 50,000 IU on day 1, day 2 and day 14

C is correct: under 6 months the dose is **50,000 IU**, and UCG 2023 gives **three doses (day 1, day 2, day 14)** to every measles case. A is wrong: 200,000 IU is for children 12 months and older; giving it to a young infant risks toxicity (bulging fontanelle, vomiting). B is wrong: wrong dose for age and a single dose is insufficient. D is wrong: infants under 6 months with measles also benefit from vitamin A. E is wrong: daily low doses are not the measles regimen.

7. A · Acute measles encephalitis: control seizures, supportive care, and do an LP (if safe) with ceftriaxone until bacterial meningitis is excluded

A is correct: reduced consciousness persisting after repeated seizures during the acute illness suggests **acute measles encephalitis** (about 1 in 1,000 cases); treat seizures (diazepam, then phenobarbital 15–20 mg/kg), give supportive care, and exclude bacterial meningitis (LP if safe, ceftriaxone meanwhile). B is wrong: two seizures with persistent drowsiness are not a simple febrile convulsion. C is wrong: SSPE occurs years later. D is wrong: her glucose is normal. E is wrong: there is no stridor.

8. E · EEG showing periodic complexes and CSF measles IgG antibody titre

E is correct: progressive cognitive decline with **periodic myoclonic jerks** years after measles in infancy is **SSPE**; diagnosis is supported by **periodic high-voltage EEG complexes** and **markedly raised CSF measles IgG** (raised CSF:serum antibody index). A is wrong: IgM indicates acute infection; SSPE shows high IgG. B is wrong: malaria does not cause a 6-month progressive decline. C is wrong: NCC causes seizures but not periodic myoclonus with dementia. D is wrong: hypothyroidism may slow cognition but does not cause periodic myoclonus.

9. B · It is given as 0.5 mL subcutaneously at 9 and 18 months, and reconstituted vaccine is discarded after 6 hours

B is correct: UNEPI gives **MR1 at 9 months and MR2 at 18 months**, **0.5 mL subcutaneously**, and reconstituted live vaccine must be discarded after **6 hours**. A is wrong: those are pentavalent/PCV times and route. C is wrong: malnutrition is an indication, not a contraindication. D is wrong: one dose at 9 months protects about 85% (maternal antibody interference); two doses about 95% or more. E is wrong: SSPE is caused by wild measles virus, not the vaccine.

10. D · Vitamin A, ReSoMal for dehydration, F-75 feeds, antibiotics per the SAM protocol, isolation and measles complications care

D is correct: oedema and MUAC <11.5 cm = **complicated SAM** with measles; manage under the **WHO 2023/UCG SAM protocol**: vitamin A, **ReSoMal** (not standard bolus fluids) for dehydration, **F-75**, routine antibiotics, warmth, glucose checks, plus isolation and measles care. A is wrong: boluses are dangerous in SAM unless there is shock. B is wrong: complicated SAM with measles and dehydration needs inpatient care. C is wrong: vitamin A is essential in measles (and SAM with eye signs). E is wrong: early feeding is part of treatment.

11. C · Roseola infantum (HHV-6)

C is correct: **roseola** typically affects children under 2 with 3–5 days of high fever, followed by a **rash appearing as the fever falls**, in a well child without the 3 Cs. A is wrong: measles has the 3 Cs, Koplik spots and fever that rises with the rash. B is wrong: rubella has mild fever with the rash and occipital/post-auricular nodes. D is wrong: scarlet fever has sore throat, sandpaper rash and strawberry tongue. E is wrong: meningococcal rash is non-blanching purpura in an ill child.

12. A · TB is still likely: measles suppresses cell-mediated immunity, reactivates TB and causes false-negative tuberculin tests; do Xpert on sputum/gastric aspirate and a chest X-ray

A is correct: measles causes **immune suppression ('immune amnesia')** that can **reactivate TB** and make the **tuberculin test falsely negative** for weeks; with a TB contact and weight loss, investigate with **GeneXpert** and **chest X-ray**. B is wrong: a negative TST never excludes TB, especially after measles. C is wrong: bronchiolitis obliterans is rare and does not cause night sweats and weight loss. D is wrong: vitamin A does not treat this. E is wrong: weight loss and night sweats are red flags, not a normal post-measles cough.

13. E · Yes: the campaign targets all children aged 9–59 months regardless of previous vaccination status; an extra dose is safe

E is correct: the MoH campaign (1–5 October 2026) aims to vaccinate **all children aged 9–59 months irrespective of previous MR status**, to close immunity gaps; extra doses of MR are safe. A is wrong: there is no harm in an extra dose, and campaigns deliberately ignore previous status. B is wrong: serology is not required. C is wrong: past measles gives immunity, but the campaign does not depend on history. D is wrong: the target age is 9–59 months.

14. B · Keep him calm with his mother, oxygen, nebulised adrenaline 5 mL of 1:1,000, and close monitoring with a plan for airway support if he worsens

B is correct: stridor at rest with recession is severe **measles croup**; keep the child calm, give **oxygen**, **nebulised adrenaline** (0.5 mL/kg of 1:1,000, max 5 mL) as a temporary measure, and monitor closely with airway support available. A is wrong: the WHO Pocket Book advises **against steroids in measles croup**, and he is too sick for discharge. C is wrong: throat examination can precipitate complete obstruction. D is wrong: salbutamol does not treat upper-airway (laryngeal) obstruction. E is wrong: lying flat and sedation can worsen obstruction and hide deterioration.

15. D · Koplik spots with a 3–4-day prodrome of cough, coryza and conjunctivitis

D is correct: **Koplik spots** and a **prodrome of the 3 Cs** before the rash are characteristic of measles. A is wrong: both cause maculopapular rashes. B is wrong: both cause fever (milder in rubella). C is wrong: both rashes start on the face and spread down. E is wrong: **post-auricular/suboccipital lymphadenopathy** is typical of **rubella**, so it points away from measles.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Measles with hypoxaemic pneumonia and malnutrition

You are the intern at a regional referral hospital paediatric ward during an outbreak in Kassanda district. A 2-year-old girl is admitted with 4 days of fever, cough, runny nose and red eyes, followed by a rash that started behind her ears 3 days ago. Her brother had a similar illness 2 weeks ago. Weight 9.5 kg, height 82 cm, MUAC 12.1 cm, no oedema.

1. What do you do first on arrival?

> **Isolate** (separate bay/room, airborne precautions, immune staff). **ETAT**: RR 54, lower chest indrawing, **SpO2 88% → oxygen** by nasal prongs to SpO2 ≥90%; HR 150, CRT 2 s (no shock); alert; no stridor; glucose 4.6. Weigh and measure: **WHZ −2.4** (moderate acute malnutrition), MUAC 12.1 cm (MAM), no oedema.

2. What history will you take quickly, and why?

> Sequence and date of rash (infectiousness until 4 days after rash onset); contacts (brother; notify the outbreak); **card**: MR1 given, **MR2 missed**; recent vitamin A/albendazole (8 months ago); feeding and **food taboos** (family stopped beans and fish); diarrhoea, ear discharge, eye pain, mouth pain; convulsions; **HIV** (mother negative; child unknown); TB contacts.

3. Examination shows clear corneas, superficial mouth ulcers, conjunctivitis without pus, no ear discharge, crackles at the right base. What are your diagnoses and treatment, with doses?

> **Measles with severe (hypoxaemic) pneumonia** and **MAM**. **Vitamin A 200,000 IU** day 1, 2 and 14 (UCG). **Ampicillin 50 mg/kg = 475 mg IV 6-hourly + gentamicin 7.5 mg/kg ≈ 70 mg IV daily** (or ceftriaxone 80–100 mg/kg/day); oxygen; paracetamol 15 mg/kg = 140 mg 6-hourly; salt-water mouth care + gentian violet 0.25%; eye cleaning; continue breastfeeding and feeds (counsel against taboos).

4. What else must you do beyond this child?

> **Notify** the district surveillance focal person (if this is among the first cases, take serum for IgM to UVRI); check vaccination of **other children on the ward** (vaccinate those aged 6 months or more without 2 doses within 72 h) and **siblings at home**; check staff immunity; health education about MR2 at 18 months.

5. On day 3 she is off oxygen, eating, afebrile. What is your discharge plan?

> Switch to **oral amoxicillin 40 mg/kg 12-hourly** to complete 5 days of antibiotics; **vitamin A day 14** dose (give a date or give to take home per policy); **HIV test**; enrol in a **supplementary feeding/MAM programme** (RUSF), growth monitoring; **MR2** once recovered; review at 1–2 weeks; warn about persistent cough or weight loss in the coming weeks (TB after measles).

Take-away: Measles kills through its complications: isolate, give age-dosed vitamin A, treat hypoxaemic pneumonia as bacterial, support nutrition, and protect every contact with MR vaccine.

Case 2: A cloudy cornea at a HC II

You are on community placement at a HC II in Kagadi district. A mother brings her 10-month-old son, who has had measles for 6 days. Since yesterday he refuses to breastfeed and keeps his eyes closed. A neighbour put 'traditional drops' in his eyes. Weight 7.4 kg.

1. What will you look for on examination?

> General danger signs (lethargy, unable to breastfeed, vomiting everything, convulsions); **eyes**: open gently (pus, **corneal clouding or ulcer**, Bitot's spots, dryness); **mouth** (depth and extent of ulcers, thrush, cheek gangrene); chest (RR, indrawing); hydration; MUAC and oedema; temperature.

2. He is lethargic, the right cornea is hazy with a central white patch, and there is pus. Classify and treat at the HC II.

> **Severe complicated measles** (danger sign + corneal clouding). Give **vitamin A 100,000 IU** now (6–11 months); **ampicillin 50 mg/kg = 370 mg IM + gentamicin 7.5 mg/kg ≈ 55 mg IM** (or benzylpenicillin 50,000 IU/kg IM at HC II); clean the eyes, apply **tetracycline 1% eye ointment**, cover the right eye with a pad; check glucose or give sugar water/breast milk by cup; **refer urgently** with a note.

3. Why is this eye an emergency?

> Measles keratitis plus **vitamin A deficiency** (and harmful traditional remedies) can progress to **corneal ulceration, keratomalacia and perforation** within days, causing permanent blindness; measles is a leading cause of childhood blindness in Africa. Vitamin A, antibiotics and an ophthalmologist are needed urgently.

4. What will the hospital do for the eye and how will you follow him up?

> Continue **vitamin A** (day 2 and day 14), topical antibiotics, eye pad, **ophthalmology review** (fluorescein staining; atropine for pain if ulcerated, per ophthalmologist); **never steroid drops**; manage any malnutrition. Follow-up: visual assessment, nutrition, and **MR vaccination for siblings**; he should still receive MR at the scheduled ages (for rubella protection; an extra measles dose is harmless).

5. How do you counsel the mother about traditional eye remedies and feeding?

> Explain gently that drops of herbs, urine or breast milk can infect or burn the eye; only clean water and the ointment from the clinic should be used. Encourage frequent breastfeeding and soft family foods including fish, eggs and beans (no food taboos); explain danger signs for return.

Take-away: Corneal clouding in measles is a blindness emergency: vitamin A, first-dose antibiotics, tetracycline ointment, an eye pad and urgent referral, and never steroid eye drops.

Case 3: A teenager with failing grades and jerks

A 14-year-old boy is referred to Mulago neurology clinic because of 6 months of declining school performance, irritability and forgetfulness. For the past month, his arms jerk every few seconds and he falls. His mother recalls that he had a 'severe measles' at 10 months, before vaccination. Weight 38 kg.

1. What is your differential diagnosis?

> **SSPE** (top), HIV encephalopathy, neurocysticercosis or other epilepsy syndromes (progressive myoclonic epilepsy), Wilson's disease, adrenoleukodystrophy, depression or substance use, CNS TB or tumour, HIV-associated opportunistic infections.

2. What examination findings support SSPE?

> **Periodic myoclonic jerks** (every 5–10 s, may cause drop attacks), cognitive impairment, ataxia, spasticity in later stages, **chorioretinitis** on fundoscopy (in some), no fever or meningism.

3. Which investigations will you request and what do you expect?

> **EEG**: periodic high-voltage slow-wave complexes (Radermecker) synchronous with jerks; **CSF**: normal cells and protein, **markedly raised measles IgG** with a raised CSF:serum antibody index (oligoclonal bands); serum measles IgG very high; **MRI**: white-matter changes; HIV test.

4. How will you counsel the family?

> Explain that SSPE is caused by the measles infection he had as a baby, that it is progressive and usually fatal within 1–3 years, that there is **no cure**, and that treatment (antiepileptics for myoclonus, isoprinosine/interferon where available, physiotherapy, nutrition, palliative care) aims at comfort. It is not contagious and is not caused by vaccines.

5. What is the public-health lesson?

> SSPE risk is highest when measles occurs **under 2 years**, before the child is fully protected; the only prevention is **high MR coverage (≥95%, both doses)** so that infants are protected by herd immunity. Uganda's MR2 coverage of 64% (2025) leaves gaps that allow infant infections.

Take-away: SSPE is a fatal late complication of wild measles in infancy, recognised by cognitive decline with periodic myoclonus, periodic EEG complexes and high CSF measles antibody, and prevented only by vaccination.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.